

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY
CHANDUBHAI S PATEL INSTITUTE OF TECHNOLOGY
DEPARTMENT OF INFORMATION TECHNOLOGY
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Subject Name: Software Engineering
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Practical List

Sr No	Aim	Hrs.
1	(Role of Software) Identify the role of the software in today's world across a few significant domains related to day to day life Background: Software has made the world a global village today. The impact of software spans across almost all aspect of human life. All organizations, Institutions and companies are leveraging the potentials of software in automating the critical functions and eliminating manual interventions. Software is also a predominant are for trade and export especially for the countries like India. Domains like health care, Airlines , financial Services, Insurance , retails, Education, and many more have exploited software and still there a lot of the scope for software to create impact and add values in multiple dimensions. Problem : In the context of this background, indentify the areas (or application or systems) Description : how software has been leveraged extensively in the following domains 1. Health Care 2. Airlines 3. Banking Insurance 4. Retail 5. Education	4
2.	(Requirement Gathering) Identify the requirement development activities associated with each of the following scenarios, a. Joe is creating an online survey questionnaire for requesting user feedback on the desired features of the application to be developed. b. Mark is preparing a formal document which includes all of the desired features identified by the survey.	2

- c. Jack identified an incomplete requirement statement.
- d. Jones is identifying all security related requirement and separating them from the performance related requirements.
- e. Merlin a team member is sent to client to observe the business case and collect typical user requirements.
- f. Leo is team member is working on requirement and ensuring that requirement collected should not be vague and unclear.
- g. Lee is conducting a facilitated meeting with the stakeholder to capture the requirements.
- h. Amit a team member is distributing questionnaires to stack holder for gathering user requirements.

Scenario	Requirement	Development	Activities
A			
B			
C			
D			
E			
F			
G			
H			

Also Design questionnaires based on Practical 1 and other tools to elicit requirements for the chosen project.

What is Project Title?

Who will use this feature?

Who will deliver the inputs for the feature?

Who will receive the outputs of the feature?

Where would the user access this feature?

What are the Hardware and Software Requirements of the Project?

What is in the frontend and backend of the application?

Who will work and how many people needed to complete Project?

What is the main aim and application of the Project?

What is the scope of the project?

What are the limitations of the project?

What is the Future Enhancement of the project?

3 (Software Development Model)

Identify the suitable (SDLC)Software development Lifecycle model for the given scenario.

Background :

“ABC” is a new enterprise recently started by a few young entrepreneurs, intended to develop robotic systems using IT based solutions. “TENZ” is a well-

	established automobile manufacturing plant, Interested in automating some of the operations of the plant. As first step towards automation, TENZ has decided to deploy a robotic arm to pick and place large number of bolts from one tray of a conveyer belt to another tray of a different conveyer belt. Moving one bolt at a time. TENZ has asked ABC for demonstration the robotic arm for the said purpose as a proof of concept. This is the first project being executed by ABC. TENZ would like to have a first experience of the robotic arm before signing the MOU (Memorandum of Understanding) with ABC. After MOU sign-off, TENZ would provide the complete set of requirement of the robot.	
4	(Requirement Classification) Problem: Classify the following requirement by selecting the appropriate option. And classify the requirement into functional and non-functional requirements for your definition Description: 1.ATM machine shall validate PIN of the user during login along with bio-metric verification. 2.“Peak transaction Volume(s)- 20,000 calls in a busy hour, average duration 20 Secs, grade of services 99.98%” 3.“Brahe System Shall sounds the alarm for 10 seconds at frequency of 100H when the brake is applied”. 4.“Mean Time to failure (MTTF) -There should be no more than three Severity-1 outage per month”.	2
5.	Design System Requirement Specification (SRS) document for the given given scenario Article Systems (AS), a giant books franchise has approached IT company ABC Ltd. To help them automate their business processes and shift them online so all of its outlets across the globe would be under one portal. AMC Ltd. , agrees to take on the project and both the companies sit down and strike a deal. After detailed discussion and analysis, ABC Ltd. decides to follows the waterfall model for development of the project. This discussion covered topics such as the business analysis and operations of AS, The expectations of the client (AS) from ABC Ltd, their needs and requirements from the software, their current existing system including its limitations and what the proposed system would do to address these issues. Taking into account all the input, the concerned team at ABC designed and developed the SRS.	2
6.	Develop a Software Project Management Plan using (SmartDraw, OpenProject, LibrePlan, ProjectLibre) tools.	2
7.	Calculate cost estimation for the project using FP calculation and COCOMO model. calculate and explore other parameters for estimation of cost of your project like : Study Analogous, Parametric, Three-point, Bottom-up estimating techniques.	2

8.	Build design documentation for given project: (UMLet 14.2 ,SmartDraw, Visio 2007) 1. Procedure oriented methodology (DFD up to level 2, structure chart) 2. Object oriented methodology (all UML diagrams) 3. GUI Design 4. Database Design 5. Navigation Design	6
9.	Design coding standards and guidelines for a given project.(C,C++,Java, HTML etc.)	2
10.	Design the following for given project: 1. Test case 2. Test Suite 3. Testing Strategy Also perform unit testing on the developed project: (http://www.testingexcellence.com/best-open-source-test-management-tools/)	2
11.	Study different CASE tools and testing tools (winrunner,load runner , QTP, qTest, IBM Rational Functional tester, MSC (message sequence chart), SDL (specification and description language),TTCN (testing and test control notation), TTCN-3 and prepare a summary report.	4

Prepare By:

Mrudang Pandya, Sanket suthar